

Lando System Specification Language v2 Grammar

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1 Notation

The grammar for the Lando System Specification Sublanguage is written in the EBNF notation. The main elements of the notation that we utilize are:

- Keyword terminals are represented in **typewriter** font.
- Single-character terminals appear in single quotes, e.g. `'!'` or `'\n'`.
- More complex terminals are represented using standard regular expressions written as `«regexp»`, which may include escaped control characters such as `\n`. Regular expressions may use character classes (with negation). Note that the pattern `«/»` matches the literal character forward slash.
- An optional *thing* is written as `[ thing ]`.
- Zero or more repetitions of *thing* are written as `{ thing }`.
- One or more repetitions of *thing* are written as `{ thing }+`.
- If *S* and *T* are non-terminals or regular expressions, *S* – *T* denotes the set of strings matching *S* but not *T*.
- *EOF* represents end-of-file; it can be thought of as a special terminal that is not consumed, but rather is replicated as often as necessary to match the grammar when parsing.
- Whitespace and line separators are significant throughout. Note that the line separator terminal $\mathcal{N}$ consumes any whitespace immediately following the actual end-of-line character(s). Two consecutive line separators (i.e. matching $\mathcal{N}\mathcal{N}$) are always lexed instead as a single blank line terminal $\mathcal{B}$; this distinction is necessary to resolve certain ambiguities in the grammar for paragraphs. The general line separator $\mathcal{L}$ is the union of $\mathcal{N}$ and $\mathcal{B}$.
- In cases of ambiguity, productions involving repetition behave “greedily,” i.e. they match as much of the input as possible. For example, because the Sentence Body production uses `{ }`, the string “**abc def ghi**” parses as a single sentence containing three words, rather than as three sentences of one word each.

1.1 Context-Free Grammar

<i>landoSource</i>	::=	$\{\mathcal{L}\}\{\text{body}\}\{\mathcal{L}\}\{\mathcal{CL}\}\{\mathcal{L}\}\text{EOF}$	<i>Lando source</i>
<i>specElement</i>	::=	$\text{system} \mid \text{subsystem} \mid \text{subsystemImport} \mid \text{component} \mid \text{componentImport}$ $\mid \text{events} \mid \text{scenarios} \mid \text{requirements} \mid \text{relation}$	<i>Top-level element</i>
<i>system</i>	::=	$\{\mathcal{CL}\}\text{system name} [\text{abbrev}] [\mathcal{C}]\{\mathcal{L}\}^+$ $\text{explanation} \{\mathcal{CL}\} [\text{indexing blockend}]$ $[\text{contains} \{\mathcal{L}\} \text{body end} [\mathcal{C}] \text{blockend}]$	<i>System</i>
<i>body</i>	::=	$\{\text{specElement}\}$	<i>Source/(Sub)system body</i>
<i>subsystem</i>	::=	$\{\mathcal{CL}\}\text{subsystem name} [\text{abbrev}]\{\text{clientClause}\} [\mathcal{C}]\{\mathcal{L}\}^+$ $\text{explanation} \{\mathcal{CL}\} [\text{indexing blockend}]$ $[\text{contains} \{\mathcal{L}\} \text{body end} [\mathcal{C}] \text{blockend}]$	<i>Subsystem</i>
<i>subsystemImport</i>	::=	$\{\mathcal{CL}\}\text{import} \{\mathcal{S}\}^+ \text{subsystem name} [\text{abbrev}]\{\text{clientClause}\} [\mathcal{C}] \text{blockend}$	<i>Subsystem import</i>
<i>explanation</i>	::=	<i>paragraph</i>	<i>Explanation</i>
<i>abbrev</i>	::=	$\{\mathcal{S}\}' (' \{\mathcal{S}\} \text{nameWord} \{\mathcal{S}\}')' \{\mathcal{S}\}$	<i>Abbreviation</i>
<i>inheritClause</i>	::=	$\text{inherit} \text{qname} \{\mathcal{L}\}\{', ' \{\mathcal{L}\} \text{qname} \{\mathcal{L}\}\}$	<i>Inherit clause</i>
<i>clientClause</i>	::=	$\text{client} \text{qname} \{\mathcal{L}\}\{', ' \{\mathcal{L}\} \text{qname} \{\mathcal{L}\}\}$	<i>Client clause</i>
<i>indexing</i>	::=	$\text{indexing} \{\mathcal{S}\}\{\{\mathcal{L}\}^+ \text{indexEntry}\}$	<i>Indexing</i>
<i>indexEntry</i>	::=	$\text{indexValue} ':' \text{indexValuePart} \{\{\mathcal{L}\}^+ \text{indexValuePart}\}$	<i>Index Entry</i>
<i>indexValuePart</i>	::=	$\text{indexValue} \{\mathcal{L}\} [\mathcal{C}]$	<i>Index Value Part</i>
<i>component</i>	::=	$\{\mathcal{CL}\}\text{component name} [\text{abbrev}]\{\text{inheritClause} \mid \text{clientClause}\} [\mathcal{C}]\{\mathcal{L}\}^+$ $\text{explanation} \{\text{componentFeature blockend}\}$	<i>Component definition</i>
<i>componentImport</i>	::=	$\{\mathcal{CL}\}\text{import} \{\mathcal{S}\}^+ \text{component name} [\text{abbrev}]\{\text{clientClause}\} [\mathcal{C}] \text{blockend}$	<i>Component import</i>
<i>componentFeature</i>	::=	$\text{constraint} \mid \text{command} \mid \text{query}$	<i>Component Feature</i>
<i>constraint</i>	::=	$\{\mathcal{CL}\}\text{sentBody} ' .' [\text{wordSep}] [\mathcal{C}]$	<i>Constraint Feature</i>
<i>query</i>	::=	$\{\mathcal{CL}\}\text{sentBody} '?' [\text{wordSep}] [\mathcal{C}]$	<i>Query Feature</i>
<i>command</i>	::=	$\{\mathcal{CL}\}\text{sentBody} '!' [\text{wordSep}] [\mathcal{C}]$	<i>Command Feature</i>
<i>events</i>	::=	$\{\mathcal{CL}\}\text{events name} [\mathcal{C}]\{\mathcal{L}\}^+ \{\text{eventEntry}\}$	<i>Events</i>
<i>eventEntry</i>	::=	$\{\mathcal{CL}\}\text{name} [\mathcal{C}]\{\mathcal{L}\}^+ \text{paragraph}$	<i>Event Entry</i>
<i>scenarios</i>	::=	$\{\mathcal{CL}\}\text{scenarios name} [\mathcal{C}]\{\mathcal{L}\}^+ \{\text{scenarioEntry}\}$	<i>Scenarios</i>
<i>scenarioEntry</i>	::=	$\{\mathcal{CL}\}\text{name} [\mathcal{C}]\{\mathcal{L}\}^+ \text{paragraph}$	<i>Scenario Entry</i>
<i>requirements</i>	::=	$\{\mathcal{CL}\}\text{requirements name} [\mathcal{C}]\{\mathcal{L}\}^+ \{\text{requirementEntry}\}$	<i>Requirements</i>
<i>requirementEntry</i>	::=	$\{\mathcal{CL}\}\text{name} [\mathcal{C}]\{\mathcal{L}\}^+ \text{paragraph}$	<i>Requirement Entry</i>
<i>relation</i>	::=	$\{\mathcal{CL}\}\text{relation qname} \{\text{inheritClause} \mid \text{clientClause}\}^+ [\mathcal{C}] \text{blockend}$	<i>Relation Declaration</i>
<i>qname</i>	::=	$\text{name} \mid \text{qname} ':' \text{name}$	<i>Name</i>
<i>name</i>	::=	$\{\mathcal{S}\} \text{nameWord} \{\{\mathcal{S}\}^+ \text{nameWord}\} \{\mathcal{S}\}$	<i>Name facet</i>
<i>nameWord</i>	::=	$\llbracket [^{\wedge} \backslash \text{r} \backslash \text{n} \backslash \text{t}():] * [^{\wedge} \backslash \text{r} \backslash \text{n} \backslash \text{t}():, . ! ?] \rrbracket - \llbracket // . * \rrbracket - \text{nonName}$	<i>Name word</i>
<i>nonName</i>	::=	$\text{client} \mid \text{inherit}$	<i>Keywords</i>
<i>indexValue</i>	::=	$\{\mathcal{S}\} \text{indexValueWord} \{\{\mathcal{S}\}^+ \text{indexValueWord}\} \{\mathcal{S}\}$	<i>Index value</i>
<i>indexValueWord</i>	::=	$\llbracket [^{\wedge} \backslash \text{r} \backslash \text{n} \backslash \text{t}] * [^{\wedge} \backslash \text{r} \backslash \text{n} \backslash \text{t}:] \rrbracket - \llbracket // . * \rrbracket$	<i>Index value word</i>
<i>paragraph</i>	::=	$\{\text{sentence}\}^+ \text{parend}$	<i>Paragraph</i>
<i>parend</i>	::=	$[\mathcal{C}](\mathcal{B}\{\mathcal{L}\} \mid \text{EOF})$	<i>Paragraph end</i>
<i>sentence</i>	::=	$\text{sentBody} [\text{sentTerm}] [\text{wordSep}]$	<i>Sentence</i>
<i>sentBody</i>	::=	$\text{sentWord} \{\text{wordSep} \text{sentWord}\} [\text{wordSep}]$	<i>Sentence Body</i>
<i>wordSep</i>	::=	$\{\mathcal{S}\}^+ \mid \{\mathcal{S}\} [\mathcal{C}] \mathcal{L}$	<i>Word Separator</i>
<i>sentWord</i>	::=	$\llbracket [^{\wedge} \backslash \text{r} \backslash \text{n} \backslash \text{t}] * [^{\wedge} \backslash \text{r} \backslash \text{n} \backslash \text{t} ! ?] \rrbracket - \llbracket // . * \rrbracket$	<i>Sentence word</i>
<i>sentTerm</i>	::=	$'.' \mid '!' \mid '?'$	<i>Sentence terminator</i>
<i>blockend</i>	::=	$\{\mathcal{L}\}^+ \mid \text{EOF}$	<i>Block end</i>
$\mathcal{S}$	::=	$\llbracket [\backslash \text{t}] \rrbracket$	<i>Whitespace character</i>
$\mathcal{N}$	::=	$['\backslash \text{r}'] '\backslash \text{n}' [\mathcal{S}] \mid '\backslash \text{r}' [\mathcal{S}] \mid \text{EOF}$	<i>Single line separator</i>

$$\begin{aligned}\mathcal{B} &::= \mathcal{NN} \\ \mathcal{L} &::= \mathcal{N} | \mathcal{B} \\ \mathcal{C} &::= \langle\langle // [^{\wedge} \backslash \mathbf{r} \backslash \mathbf{n}] * \rangle\rangle\end{aligned}$$

Blank line
General line separator
Comment

2 Context-Sensitive Considerations

The following extra-grammatical rules are applied during or after parsing the concrete grammar above.

- If a *system* or *subsystem* element does not have an explicit body, then it has an implicit body that begins immediately after the element and extends until just before the next **system**, **subsystem** or unmatched **end**.
- *subsystemImport* can only appear in an (explicit or implicit) *system* or *subsystem* body. *component* and *componentImport* can only appear in an (explicit or implicit) *subsystem* body.
- A *sentence* that does not end with a '.', '!', or '?' symbol (possibly followed by white space characters or line separators) induces a warning message.
- Note that because of the “greedy” resolution of ambiguity, this warning can only occur at the end of a paragraph.